

EXPERNETIC

Talent Assessment Program · Data Engineer Intern

FINAL REPORT

Inside Airbnb Market Intelligence Platform

Data Engineering, Analytics & AI Project

Prepared by

Thamindu Weerasinghe

Submitted to

Expernetic (Pvt) Ltd — Data Engineer Intern Technical Assessment

Market: Amsterdam, North Holland, The Netherlands

Data Snapshot: 15 June 2026

Report Date: 12 July 2026

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1. Executive Summary

This project builds an end-to-end Airbnb market intelligence platform using Inside Airbnb data for Amsterdam, North Holland, The Netherlands.

The project transforms raw Airbnb datasets into cleaned analytical datasets, validates data quality, builds a DuckDB warehouse, performs exploratory and statistical analysis, and trains a machine learning model to predict listing prices.

The final solution demonstrates data engineering, analytics, statistical reasoning, machine learning, responsible AI usage, and professional project documentation.

Key Outcomes

- Built a reusable data pipeline under `src/`
- Created data quality validation reports
- Built a final listing master dataset
- Built a DuckDB analytical warehouse with star schema design
- Created SQL analytical summary tables
- Completed EDA and business analysis
- Completed statistical testing with effect size interpretation
- Trained and evaluated multiple ML models for price prediction
- Selected Random Forest as the final model
- Documented AI usage, assumptions, limitations, and open innovation opportunities

2. Project Objective

The objective of this project is to convert raw Airbnb market data into a structured market intelligence platform.

The platform aims to answer business questions such as:

- Which room types dominate the Amsterdam Airbnb market?
- Which neighbourhoods show stronger market activity?
- How do prices vary by room type, host type, and neighbourhood?
- What features are associated with listing price?
- Can machine learning estimate Airbnb listing prices?
- Where does the model perform well or struggle?
- How can AI be used to make market intelligence easier for business users?

3. Dataset Overview

The project uses Inside Airbnb data for:

- City: Amsterdam, North Holland, The Netherlands
- Snapshot date: 2026-06-15

Raw Files Used

- listings.csv.gz
- calendar.csv.gz
- reviews.csv.gz
- listings.csv
- reviews.csv
- neighbourhoods.csv
- neighbourhoods.geojson

Main Dataset Roles

Dataset	Purpose
listings.csv	Summary listing information
listings.csv.gz	Detailed listing attributes
calendar.csv.gz	Availability and stay-policy metrics
reviews.csv.gz	Detailed review activity
reviews.csv	Summary review dates
neighbourhoods.csv	Neighbourhood lookup
neighbourhoods.geojson	Geospatial reference

Important dataset limitation: The Amsterdam calendar file does not contain daily price or adjusted price fields. Therefore, calendar data is used for availability and occupancy proxy analysis, not actual daily revenue analysis.

4. Project Architecture

The project follows this architecture:

```
Raw Inside Airbnb Data | v Data Ingestion | v Dataset Profiling and Data
Quality Validation | v Cleaning and Aggregation | v Final Listing Master
Table | v DuckDB Analytical Warehouse | v EDA, SQL Analytics, Statistical
Analysis, and Machine Learning | v Reports, Figures, Model Outputs, and Business
Insights
```

The reusable pipeline can be executed using:

```
python -m src.pipeline_runner
```

The machine learning workflow can be executed using:

```
python -m src.ml.train_price_model
```

5. Data Engineering Work

The data engineering workflow includes:

- Raw file validation
- Dataset inventory generation
- Schema profiling
- Missing value profiling
- Data quality checks
- Listings cleaning
- Calendar cleaning and aggregation
- Reviews cleaning and aggregation
- Final listing master creation
- DuckDB warehouse build
- Star schema design
- Reusable pipeline runner

Main Reusable Scripts

- src/ingestion/raw_data_loader.py
- src/profiling/dataset_profiler.py
- src/validation/data_quality_checks.py
- src/cleaning/listings_cleaner.py
- src/cleaning/calendar_cleaner.py
- src/cleaning/reviews_cleaner.py
- src/transformation/listing_master_builder.py
- src/warehouse/duckdb_builder.py

- `src/ml/train_price_model.py`
- `src/pipeline_runner.py`

This improves reproducibility because the project is not only notebook-based. The key workflow can be rerun through scripts.

6. Data Quality Validation

The project validates important data quality rules, including:

- Missing listing IDs
- Duplicate listing IDs
- Missing and invalid prices
- Invalid latitude and longitude values
- Calendar availability values
- Duplicate listing-date calendar rows
- Review ID uniqueness
- Relationship coverage between listings, calendar, and reviews
- Neighbourhood lookup coverage

Important Quality Findings

- Summary listings contain 10,465 unique listing rows.
- The calendar file contains 3,819,725 rows.
- The detailed reviews file contains 545,162 rows.
- 3,994 listings have missing or invalid price values.
- 6,471 listings have valid positive prices for ML.
- Neighbourhood coverage is complete for the Amsterdam neighbourhood lookup.

7. Final Listing Master

The final listing master combines:

- Cleaned summary listings
- Detailed listing features
- Calendar availability metrics
- Review activity metrics
- Host portfolio segments
- Availability segments
- Neighbourhood-level market metrics

Expected Final Listing Master Summary

Metric	Value
Listing rows	10,465
Unique listing IDs	10,465
Valid price rows	6,471
Missing or invalid price rows	3,994
Listings with calendar metrics	10,465
Listings with review metrics	Around 9,432
Neighbourhood count	22

8. Warehouse Design

The project uses DuckDB as the analytical warehouse.

Warehouse file:

```
warehouse/airbnb_market.duckdb
```

The warehouse includes a star schema.

Dimension Tables

- dim_listing
- dim_host
- dim_neighbourhood

Fact Table

- fact_listing_market

Analytical Summary Tables

- market_overview
- room_type_summary
- neighbourhood_summary
- host_portfolio_summary
- review_score_summary

The fact table grain is: one row per Airbnb listing.

The warehouse supports business analysis by room type, host segment, neighbourhood, review score, price, availability, occupancy proxy, and estimated revenue proxy.

9. Key Metrics

Availability Rate

$$\text{availability_rate} = \text{available_days} / \text{calendar_days}$$

This measures the proportion of calendar days where a listing is marked available.

Occupancy Proxy

$$\text{occupancy_proxy} = \text{unavailable_days} / \text{calendar_days}$$

This is only a proxy. Unavailable days may include booked dates or dates blocked by the host.

Estimated Revenue Proxy

$$\text{estimated_revenue_proxy} = \text{price_numeric} \times \text{unavailable_days}$$

This is not actual revenue. It is an estimated proxy because confirmed booking revenue is not available.

10. Exploratory Data Analysis

EDA was performed to understand Airbnb market structure and business patterns.

Main EDA Areas

- Market overview
- Room type analysis
- Neighbourhood analysis
- Host portfolio analysis
- Review score analysis
- Price distribution
- Availability and occupancy proxy
- Correlation analysis

Key Findings

- Entire homes/apartments dominate Amsterdam Airbnb supply.
- Price is strongly right-skewed, so median price is more reliable than average price.
- Neighbourhoods show strong differences in supply, price, review activity, and estimated revenue proxy.
- Review scores are concentrated near the high end, suggesting review score inflation.
- High-price listings and unusual properties create more prediction difficulty later in ML.
- Availability and occupancy proxy should be interpreted carefully because unavailable days are not confirmed bookings.

11. Statistical Analysis

The project used non-parametric statistical methods because Airbnb price and review score distributions are skewed and contain outliers.

Methods Used

- Kruskal-Wallis test
- Mann-Whitney U pairwise comparisons
- Bonferroni correction
- Effect size interpretation
- Spearman correlation analysis

Statistical Questions Tested

- Do prices differ significantly by room type?
- Do prices differ by host portfolio segment?
- Does occupancy proxy differ by room type?
- Do review scores differ by room type?
- Which numeric features are associated with price?

Important Findings

- Room type price differences are statistically significant.
- Host portfolio segment price differences are statistically significant.
- Occupancy proxy differs by room type.
- Review score distributions differ by room type.
- Property size features such as bedrooms, beds, bathrooms, and accommodates are meaningfully associated with price.
- Availability rate and occupancy proxy have weaker direct correlation with price.

The statistical work confirms that several EDA patterns are not only visual patterns but are supported by statistical testing.

12. Machine Learning

The project builds a supervised regression model to estimate Airbnb listing price.

Target variable: price_numeric

Rows used for ML: 6,471 listings with valid positive prices

Train/Test Split

- 80% training
- 20% testing

Models Compared

- Dummy Median Baseline
- Linear Regression
- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Final selected model: Random Forest Regressor

Final Model Performance

Metric	Value
MAE	88.40
RMSE	147.30
R ² Score	0.5009

Baseline Comparison

Metric	Value
Dummy Median Baseline RMSE	213.50
RMSE Improvement	66.20
RMSE Improvement Percentage	31.01%

The Random Forest model reduced RMSE by 31.01% compared with the dummy baseline. This shows that listing, host, review, availability, and neighbourhood features contain useful signals for price prediction.

13. Model Explainability

Feature importance analysis shows that the model uses these features strongly:

- Bedrooms
- Review scores value
- Minimum nights
- Beds
- Bathrooms
- Neighbourhood context
- Average reviews per year
- Amenities count
- Reviews per month
- Accommodates
- Room type
- Availability and occupancy proxy

Important Interpretation

Feature importance shows predictive value, not causation.

For example, bedrooms are a strong predictive signal for price, but this does not mean bedrooms alone cause price changes.

14. Residual and Error Analysis

Residual analysis was used to understand where the model makes mistakes.

Residual Summary

Metric	Value
Mean Error	-9.84
Median Error	-19.76
Mean Absolute Error	88.40
Median Absolute Error	58.41
Maximum Absolute Error	1,578.65
Prediction Count	1,295

The negative mean error shows that the model slightly overpredicts on average.

Large errors are mostly found in unusual or premium listings, such as:

- Luxury houseboats
- Historical sailing ships
- Canal-view properties
- High-end townhouses
- Unique premium properties

This is expected because the dataset does not fully capture photos, interior quality, exact view, luxury design, or street-level desirability.

15. AI / LLM Usage

AI tools were used responsibly as support during the project.

AI Assistance Was Used For

- Project planning
- Debugging
- Explaining technical concepts
- Improving documentation
- Interpreting statistical and ML outputs
- Designing reusable pipeline structure
- Preparing report and README content

AI was not used to fabricate results.

All important outputs were checked against:

- Actual dataset schemas
- Notebook outputs
- Terminal outputs
- Generated CSV reports
- Model metrics
- Feature importance outputs
- Residual analysis outputs

The project follows a human-in-the-loop approach.

16. Open Innovation Concept

A future extension of this project is an:

AI-powered Airbnb Market Intelligence Assistant

This assistant could allow business users to ask natural language questions such as:

- Which neighbourhoods have the highest estimated revenue proxy?
- Which room types perform best?
- Why is a listing predicted to be expensive?
- Which features influence price?
- Where does the model struggle?
- What are the risks of interpreting occupancy proxy?

The Assistant Could Use

- DuckDB warehouse
- SQL query templates
- Generated reports
- ML outputs
- LLM-based explanation generation
- Guardrails for proxy metrics and limitations

Important Guardrails

- Do not treat estimated revenue proxy as actual revenue.
- Do not treat occupancy proxy as confirmed occupancy.
- Do not make unsupported claims about hosts or guests.
- Always mention limitations when using proxy metrics.
- Use database outputs instead of guessing.

17. Key Assumptions

Important project assumptions:

- Summary listings are used as the base listing universe.
- Calendar unavailable days are used as an occupancy proxy.
- Estimated revenue proxy is not actual revenue.
- Missing prices reduce price analysis and ML coverage.
- Neighbourhood metrics are useful for market context.
- One snapshot is not enough to make time-series conclusions.
- ML predictions are approximate and not final pricing recommendations.

18. Key Limitations

This project has the following limitations:

- Only one city snapshot is analyzed.
- Calendar data does not include confirmed bookings.
- Calendar data does not include daily price fields.
- Estimated revenue proxy is not actual revenue.
- Some listings have missing prices.
- ML does not include photos, interior quality, exact view, events, or seasonality.
- Some small room type groups have limited sample sizes.
- Results may not generalize to other cities or future dates.

19. Business Value

This project provides business value by turning raw Airbnb data into structured market intelligence.

Potential Users Could Include

- Data analysts
- Market researchers
- Property investors
- Tourism analysts
- Hosts
- Business intelligence teams

The Platform Helps Users Understand

- Market supply
- Price patterns
- Neighbourhood performance
- Host behavior
- Review activity
- Availability behavior
- Price prediction drivers
- Model limitations

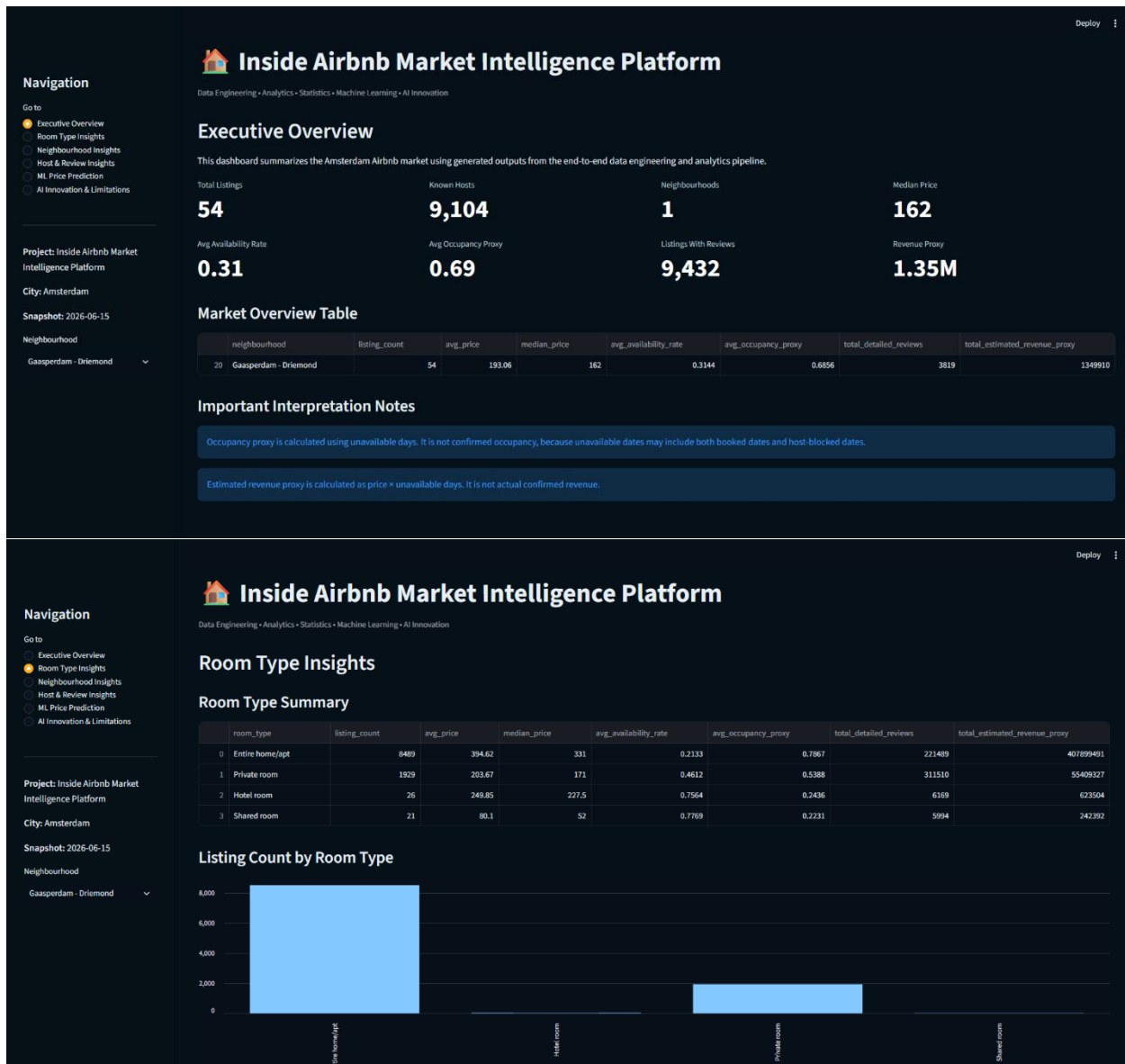
20. Business Recommendations

- Prioritize entire-home/apartment analysis because this segment dominates supply and revenue proxy.
- Use median price instead of average price for pricing decisions because price is right-skewed.

- Treat high-revenue-proxy neighbourhoods as priority areas, but validate with supply and price distribution.
- Use review score and review frequency carefully because review scores are inflated near the high end.
- Use the ML model as a decision-support tool, not an automated pricing system.
- Improve future analysis with confirmed booking data, daily prices, and multi-city snapshots.

21. Streamlit Dashboard Screenshots

This section presents visual evidence of the Streamlit analytics dashboard built for this project.



Deploy

Neighbourhood Summary

	neighbourhood	listing_count	avg_price	median_price	avg_availability_rate	avg_occupancy_proxy	total_detailed_reviews	total_estimated_revenue_proxy
6	Oud-Oost	670	308.24	276	0.3418	0.7583	25575	25874801
7	Oud-Noord	479	357.02	266	0.2248	0.7752	25295	19369930
8	Bos en Lommer	546	281.79	265	0.1675	0.8325	17223	17675443
9	Oostelijk Havengebied - Indische B	431	290.24	262	0.2123	0.7877	16186	14748957
10	Noord-West	312	265.23	259	0.2008	0.7992	10339	14127902
11	Watergraafmeer	304	311.24	280	0.2552	0.7448	10560	13291645
12	Uitburg - Zeeburgereiland	187	330.72	287	0.2342	0.7658	8494	8675081
13	Noord-Oost	179	288.65	266.5	0.2646	0.7354	7073	8353883
14	Slotervaart	223	253.38	233	0.2283	0.7717	10228	6851824
15	Geuzenveld - Slotemeer	141	250.29	233	0.2782	0.7218	5544	5469557

Select number of neighbourhoods

10

Top Neighbourhoods by Estimated Revenue Proxy

Deploy

Inside Airbnb Market Intelligence Platform

Data Engineering • Analytics • Statistics • Machine Learning • AI Innovation

Host & Review Insights

Host Portfolio Summary

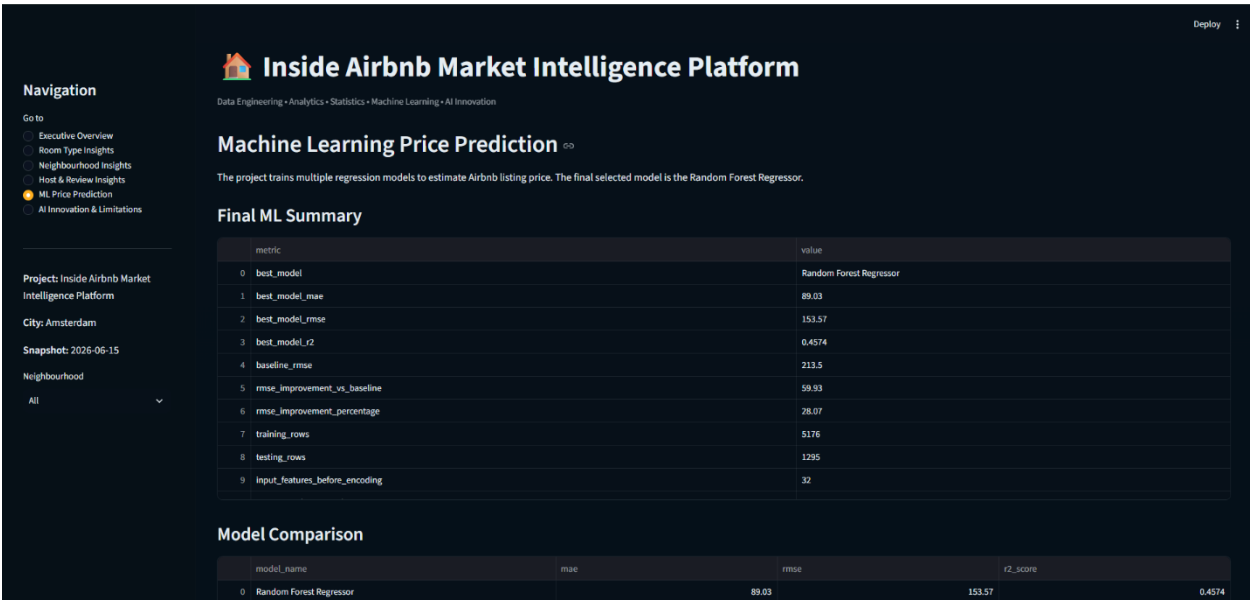
	host_portfolio_segment	listing_count	host_count	avg_price	median_price	avg_availability_rate	avg_occup
0	Single-listing Host	8509	8509	371.11	318	0.2085	
1	Small Portfolio Host	1309	537	274.87	202	0.4396	
2	Medium Portfolio Host	483	55	276.4	209	0.564	
3	Unknown Host	96	0	177.53	141	0.7595	
4	Large Portfolio Host	68	3	189.39	151	0.6113	

Review Score Summary

	review_score_band	listing_count	avg_price	median_price	avg_availability_rate	avg_occupancy_proxy	total
0	Excellent	7047	356.29	309	0.2211		0.7789
1	Very good	1794	292.38	241	0.3018		0.6982
2	No rating	1119	389.79	286	0.3778		0.6222
3	Good	434	250.67	204	0.4136		0.5864
4	Below 4.0	71	465.57	212	0.4886		0.5114

Bar chart showing host portfolio summary. The y-axis ranges from 0 to 2,000. The x-axis lists host portfolio segments. The bars are blue. The highest count is for Single-listing Host at approximately 8,500.

Bar chart showing review score summary. The y-axis ranges from 0 to 8,000. The x-axis lists review score bands. The bars are blue. The highest count is for Excellent at approximately 7,000.



22. Reflection and Prioritization

This assignment was intentionally broad, so I prioritized depth over breadth. I selected one city, Amsterdam, and focused on building a complete end-to-end workflow: ingestion, profiling, validation, cleaning, warehouse design, SQL analytics, EDA, statistical analysis, ML, dashboarding, and documentation.

I did not attempt multi-city analysis, Docker, cloud deployment, or a full RAG assistant because I prioritized reproducibility, data quality, and business interpretation for the selected city.

23. Completed and Incomplete Work Summary

Area	Status
Dataset familiarization	Completed
Data pipeline	Completed
DuckDB warehouse	Completed
EDA	Completed
Statistical analysis	Completed
ML price prediction	Completed
Streamlit dashboard	Completed
AI usage disclosure	Completed
Multi-city analysis	Not attempted
Docker	Not attempted

Area	Status
Cloud deployment	Not attempted
Full RAG assistant	Future work

24. Professionalism and Reproducibility

The repository is structured professionally with:

- Modular source code
- Reusable pipeline runner
- Data documentation
- Warehouse documentation
- Model card
- Rubric alignment
- AI usage disclosure
- Assumption and decision logs
- .gitignore for large generated files
- Clear setup and run instructions

Reproducibility Commands

```
python -m src.pipeline_runner python -m src.ml.train_price_model
```

Generated artifacts such as raw data, processed parquet files, DuckDB database files, and model files are not committed to Git because they can be recreated.

25. Rubric Coverage Summary

Competency	Coverage
Data Engineering Fundamentals	Strong
Analytical Thinking	Strong
Statistical Reasoning	Strong
Data Science & ML	Strong
AI & LLM Literacy	Good
Communication & Storytelling	Strong after documentation
Creativity & Initiative	Strong through end-to-end platform and AI assistant concept
Professionalism	Strong through modular code and reproducibility

26. Final Conclusion

This project demonstrates a complete end-to-end data workflow using real-world Airbnb market data.

It begins with raw data ingestion and profiling, then continues through cleaning, validation, transformation, warehouse design, SQL analytics, EDA, statistical testing, machine learning, explainability, and responsible AI documentation.

The final solution is more than a notebook analysis. It is a reproducible market intelligence platform with reusable source code, documented assumptions, analytical outputs, ML results, and a clear path for future AI-powered extension.

Appendix A — AI Usage Disclosure

This appendix documents the AI tools used during the project, the sections where AI assistance was used, key prompts, validation methods, and how AI-generated outputs were reviewed and modified.

A.1 AI Tools Used

The following AI tools were used during this assignment:

- ChatGPT
- Codex / AI coding assistant
- Claude AI
- GitHub Copilot or similar coding assistance, where applicable

AI tools were used as support for planning, learning, debugging, documentation, and review. The final analysis, validation, interpretation, and submission decisions remained the responsibility of the project owner.

A.2 AI-Assisted Sections

AI assistance was used in the following project areas:

Area	AI Assistance
Project planning	Breaking the assignment into clear phases and prioritizing work
Data engineering	Planning ingestion, profiling, validation, cleaning, and transformation workflows
Debugging	Understanding and fixing Python, pandas, DuckDB, scikit-learn, and Streamlit issues
Statistical analysis	Selecting suitable statistical tests and interpreting results
Machine learning	Planning model comparison, target leakage prevention, feature importance, and residual analysis

Area	AI Assistance
Documentation	Drafting README, final report, model card, architecture documentation, and rubric alignment
Dashboard development	Improving Streamlit dashboard layout, UI structure, and visual storytelling
Final review	Comparing the project against the official assessment requirements

A.3 Key Prompts Used

Prompt 1 — Project Planning

I am doing a Data Engineer Intern technical assessment using Inside Airbnb data. Help me create a step-by-step project plan that covers dataset familiarization, data engineering, EDA, statistics, ML, AI usage, documentation, and final submission.

Purpose: To create a structured project roadmap and avoid approaching the assignment randomly.

Validation: The plan was compared against the official assignment requirements and updated as the project progressed.

Prompt 2 — Dataset Familiarization

Guide me step by step to inspect the Inside Airbnb files. I need to understand the schema, row counts, columns, missing values, relationships, and dataset limitations before building the pipeline.

Purpose: To follow a schema-first approach before engineering or analysis.

Validation: Actual notebook outputs were used to confirm file shapes, columns, missing values, and relationships.

Prompt 3 — Data Cleaning and Pipeline Design

Help me clean the Airbnb listings, calendar, and reviews datasets. Explain each step clearly and help me create reusable cleaning logic without assuming columns that are not actually present.

Purpose: To design safe and reusable cleaning logic.

Validation: Cleaned outputs were checked using row counts, duplicate checks, null checks, validation reports, and relationship coverage checks.

Prompt 4 — Calendar Limitation

The Amsterdam calendar file does not contain price or adjusted_price columns. How should I handle revenue and occupancy analysis honestly?

Purpose: To avoid incorrect revenue calculations.

Validation: The final project documents that calendar data is used for availability and occupancy proxy analysis only. Estimated revenue proxy is clearly documented as a proxy, not actual revenue.

Prompt 5 — DuckDB Warehouse Design

Help me design a DuckDB analytical warehouse and star schema for the final Airbnb listing master table. Include dimension tables, fact table, analytical summary tables, and decision trade-offs.

Purpose: To design a structured analytical model suitable for SQL analytics and business reporting.

Validation: Warehouse table counts, joins, and analytical outputs were checked after building the DuckDB database.

Prompt 6 — Statistical Analysis

Help me choose appropriate statistical tests for Airbnb price, occupancy proxy, and review score comparisons. The distributions are skewed and contain outliers, so explain why non-parametric tests may be better.

Purpose: To select suitable statistical methods.

Validation: Kruskal-Wallis tests, Mann-Whitney U tests, Bonferroni correction, effect sizes, and Spearman correlations were executed in the statistical analysis notebook.

Prompt 7 — Machine Learning Price Prediction

Guide me through building a price prediction model using the final listing master table. Avoid target leakage, compare multiple models, evaluate using MAE, RMSE, and R^2 , and explain feature importance and residual errors.

Purpose: To build a structured and explainable ML workflow.

Validation: Model results were checked using train/test split outputs, model comparison metrics, feature importance, and residual analysis.

Prompt 8 — Streamlit Dashboard Improvement

You are a senior Streamlit UI/UX engineer and data visualization specialist. Improve my working dashboard into a premium, modern, professional analytics dashboard without changing the data logic, file paths, KPI calculations, or business meaning.

Purpose: To improve dashboard presentation and communication quality.

Validation: The dashboard was run locally, checked visually, and reviewed to ensure data logic and business meaning were preserved.

Prompt 9 — Final Rubric Review

Compare my final project against the official assignment rubric and identify what is completed, what is missing, and what final documents I need before submission.

Purpose: To ensure the final repository and report align with the assessment requirements.

Validation: The final checklist was compared with the official assignment PDF and remaining gaps were addressed.

A.4 Output Validation Notes

AI-generated outputs were validated using:

- Actual dataset schemas
- Notebook outputs
- Terminal outputs
- Data quality reports
- Row count checks
- DuckDB warehouse validation
- SQL query outputs
- Statistical test results
- Model evaluation metrics
- Feature importance outputs
- Residual analysis outputs
- Dashboard execution and visual review

No major result or conclusion was accepted only because AI suggested it. Outputs were checked against executed code and generated project artifacts.

A.5 Modifications Made to AI Outputs

Several AI suggestions were modified before final use.

AI Suggestion	Modification Made
Calendar revenue analysis	Adjusted because the calendar file did not contain daily price fields
Estimated revenue logic	Documented as a proxy, not actual revenue
ML features	Removed target leakage features such as estimated_revenue_proxy
Dashboard UI	Refined using manual review and Codex support
Documentation	Edited to match actual project outputs and limitations
Final report	Expanded to include prioritization, completed/incomplete work, dashboard evidence, and AI disclosure

A.6 Rejected or Limited AI Suggestions

Some AI suggestions were not implemented because they were outside the prioritized scope.

Suggestion	Reason
Multi-city analysis	One city was prioritized for deeper engineering quality and storytelling
Actual daily revenue analysis	Calendar file did not contain daily price data
Full RAG assistant	Too large for the available timeframe; documented as future work
Dockerization	Reusable local pipeline was prioritized first
SHAP/LIME explainability	Random Forest feature importance and residual analysis were completed first
Cloud deployment	Local reproducibility and GitHub submission were prioritized

A.7 Final AI Usage Statement

AI tools were used responsibly as support during the project. They helped with planning, debugging, explanation, documentation, and review.

However, the final project is based on executed code, actual generated outputs, manual validation, and documented assumptions. AI was not used to fabricate results, and all important conclusions were reviewed before inclusion in the final submission.